

Freight Rate Prediction

Data split

The labeled train-test data contained 48,000 freight-rate records from the first ten months of 2025. The final validation data covered November and December. Furthermore, the fixed December chart was clearly intended to assess the model's capability to predict freight rates in the future. As such, it was concluded that the model should specifically be strong in predicting rows from future dates.

Since the final prediction period covered two months, the last two months of the labeled data were selected as the internal validation period:

- **Training data:** January 1 to August 31, 2025 (38,477 rows)
- **Internal validation data:** September 1 to October 31, 2025 (9,523 rows)
- **Final prediction data:** November 1 to December 31, 2025 (12,000 unlabeled rows)

Choosing September and October as internal validation data allowed the model's capability to predict later dates to be evaluated more accurately. A non-chronological split might have created better-looking results, but it would have allowed later market conditions to influence a model evaluated on earlier rows.

During evaluation, all preprocessing rules and model parameters were fitted using only the January-August training data. September and October were evaluated separately and together. Model changes were accepted only when they improved both months rather than improving one month at the expense of the other.

After the preprocessing and model configuration had been selected, the final model was trained again using all 48,000 labeled rows. This allowed it to use the most recent available information before producing the 12,000 final predictions. The reported metrics still come from the January-August model predicting the unseen September-October period.

Data findings and preprocessing

Before training any models, the train-test data, validation data and December chart data were evaluated. The following was observed:

- A number of weights were negative. It was assumed that merely the sign was corrupted and, as such, the absolute value of all weights was used. Removing those rows instead of fixing their signs was also considered. It appeared that fixing the signs, as opposed to removing the rows altogether, did not reduce performance.

- Some weight and market-index values were missing. These values were replaced using medians learned from the applicable training data.
- Eight new cities appeared in the final validation data. The supplied validation rows already contained their coordinates, so those values could be used directly. Unknown categorical values were handled by the fitted one-hot encoder.
- Adding the real coordinates of the cities and the real distance between them was considered. While checking the coordinates, it became apparent that the supplied coordinates were in many cases nowhere near the real coordinates of the named cities.
- Existing coordinate values were never overwritten. When coordinates were missing, coordinates previously observed for the same city in the labeled training data were used where available.
- Considering that routes could have different levels of traffic on and before holidays and weekends, and that working during such times might require more pay, US federal holidays, weekdays and their proximity to each date were added as features.
- It was observed that `quote_signal` generally represented the rate per mile for each row, although with considerable error. This information was later used while developing the model, while the remaining features allowed the model to correct some of the errors in the supplied signal.
- The market index showed a close relationship with the date. When it was missing, the median market-index value learned from the applicable training data was used.
- The fixed December chart did not contain coordinates, `quote_signal` or `market_index`. The coordinates were retrieved from previously observed coordinates for the same cities. The missing quote signal and market index were replaced with medians learned from the labeled training data.
- The effectiveness of this approach was tested by hiding both signals in the September-October internal validation data. Combined MAE increased from \$87.34 with the signals available to \$99.47 with both signals hidden. As such, the model still maintained reasonable performance without inventing synthetic market values.
- All categorical features, including pickup city, delivery city, equipment type and weekday, were provided to the model using one-hot encoding.
- The model predicted rate per mile rather than the total posted rate directly. The predicted rate per mile was then multiplied by distance to produce the final predicted load rate.
- Around 140 rows in the internal validation data produced extreme errors. Some appeared severely overpriced and others severely underpriced relative to the available features. Neither the tested models nor `quote_signal` could accurately predict these cases. This

suggested that the rates may have been influenced by a hidden factor not presented in the dataset. However, there was not enough evidence to conclude that the target values were corrupted. As such, the rows were treated as real observations and were not removed.

Results

The following metrics were calculated using the chronological September-October internal validation period.

Metric	Training data	September 2025 testing data	October 2025 testing data	Combined testing data
MAE (\$)	68.777	79.411	94.991	87.351
Median AE (\$)	13.658	17.851	23.478	20.395
RMSE (\$)	564.987	611.303	647.004	629.749
R2	0.854	0.839	0.821	0.83
Bias (\$)	-20.162	-40.783	-56.574	-48.83
MAPE (%)	3.22	3.221	4.404	3.824
WAPE (%)	2.903	3.3	3.993	3.651
P90 error (\$)	52.816	71.031	86.291	78.945
Within \$100 (%)	97.097	94.732	92.603	93.647
Within 10% (%)	98.615	98.63	98.352	98.488

The primary model-selection metric was the combined September-October mean absolute error in dollars. The separate monthly results were also considered to ensure that improvements were consistent across both months.

Fixed December prediction chart

The following fixed-lane December prediction chart was produced by the provided score.py script after predicting every row in december_chart_inputs.csv.

